

Лабораторная работа № 9. Использование протокола STP. Агрегирование каналов

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Цель работы

- ▶ Изучение возможностей протокола STP и его модификаций по обеспечению отказоустойчивости сети.
- ▶ Агрегирование интерфейсов и перераспределение нагрузки между ними.

Топология сети

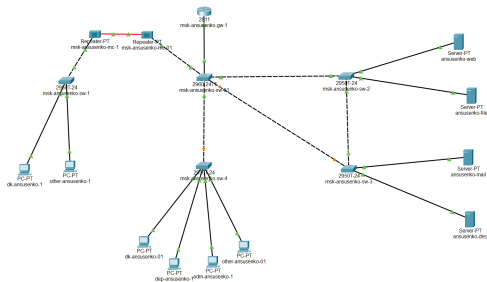
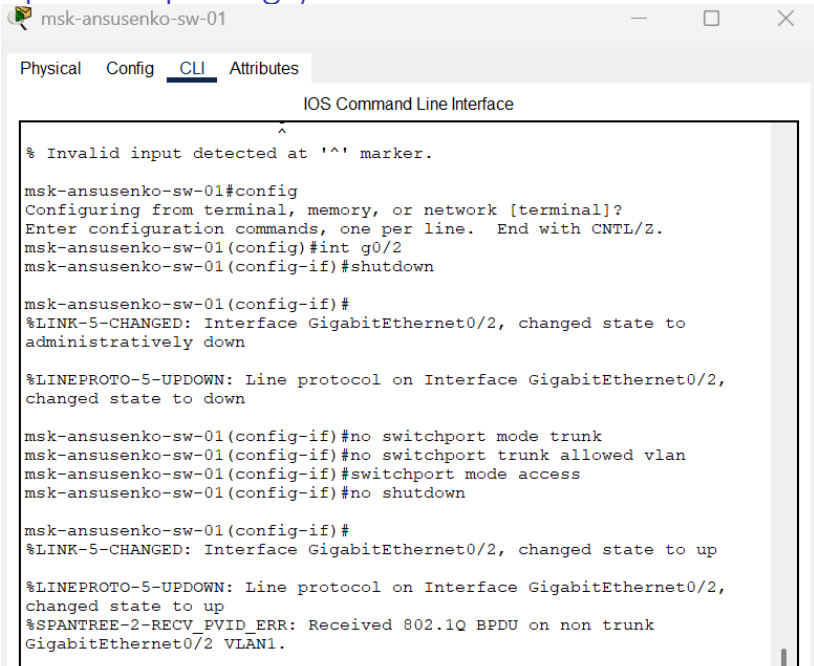


Рис.: Обновлённая топология сети.

Настройка порта Gig0/2 на msk-ansusenko-sw-01



```
Physical  Config  CLI  Attributes

IOS Command Line Interface

% Invalid input detected at '^' marker.

msk-ansusenko-sw-01#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
msk-ansusenko-sw-01(config)#int g0/2
msk-ansusenko-sw-01(config-if)#shutdown

msk-ansusenko-sw-01(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to
administratively down

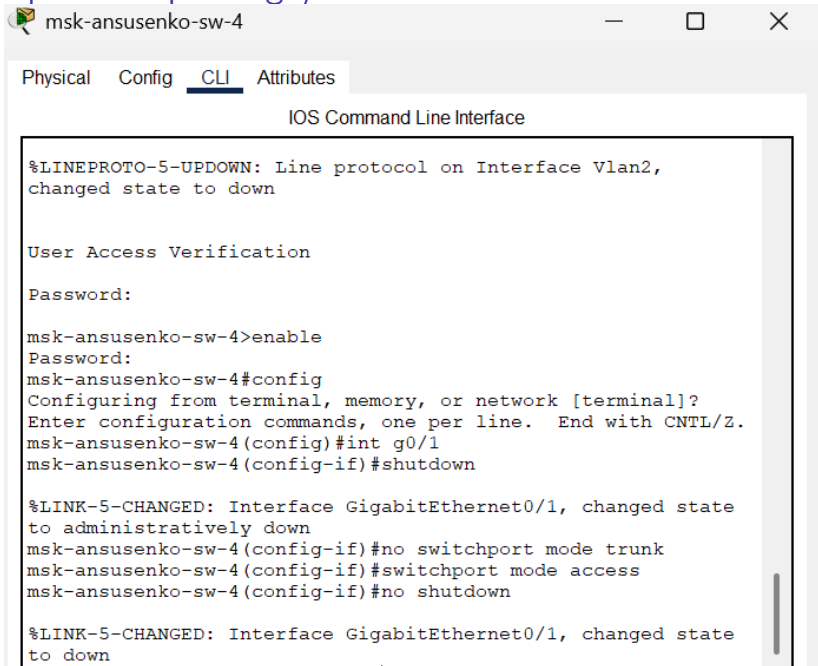
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to down

msk-ansusenko-sw-01(config-if)#no switchport mode trunk
msk-ansusenko-sw-01(config-if)#no switchport trunk allowed vlan
msk-ansusenko-sw-01(config-if)#switchport mode access
msk-ansusenko-sw-01(config-if)#no shutdown

msk-ansusenko-sw-01(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
GigabitEthernet0/2 VLAN1.
```

Настройка порта Gig0/1 на msk-ansusenko-sw-4



The screenshot shows a terminal window for a network device named 'msk-ansusenko-sw-4'. The window has tabs for 'Physical', 'Config', 'CLI' (which is selected), and 'Attributes'. The title bar includes standard window controls (minimize, maximize, close). The main content area is titled 'IOS Command Line Interface' and displays the following text:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2,
changed state to down

User Access Verification

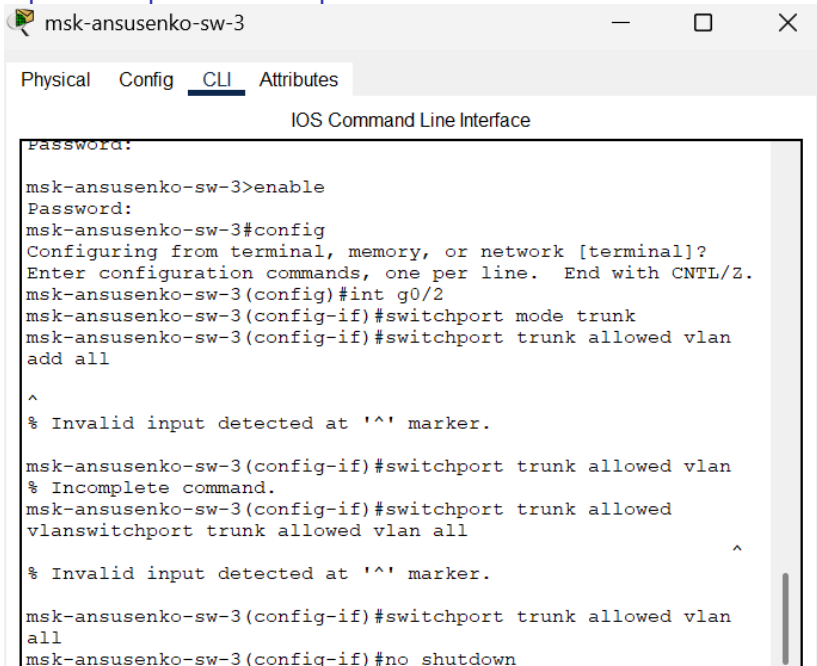
Password:

msk-ansusenko-sw-4>enable
Password:
msk-ansusenko-sw-4#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
msk-ansusenko-sw-4(config)#int g0/1
msk-ansusenko-sw-4(config-if)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state
to administratively down
msk-ansusenko-sw-4(config-if)#no switchport mode trunk
msk-ansusenko-sw-4(config-if)#switchport mode access
msk-ansusenko-sw-4(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state
to down
```

Настройка транкового режима

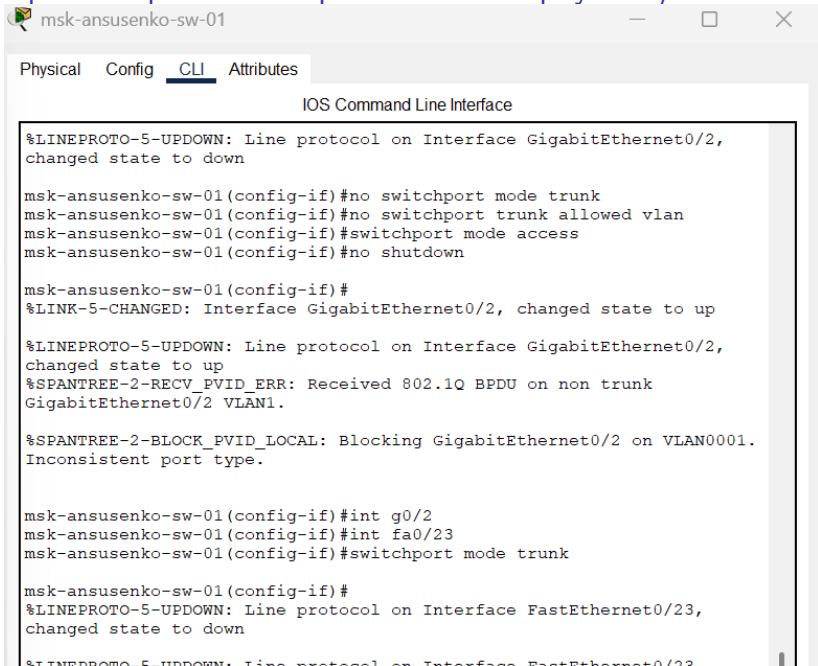


```
msk-ansusenko-sw-3
Physical Config CLI Attributes
IOS Command Line Interface
Password:
msk-ansusenko-sw-3>enable
Password:
msk-ansusenko-sw-3#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-ansusenko-sw-3(config)#int g0/2
msk-ansusenko-sw-3(config-if)#switchport mode trunk
msk-ansusenko-sw-3(config-if)#switchport trunk allowed vlan
add all
^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config-if)#switchport trunk allowed vlan
% Incomplete command.
msk-ansusenko-sw-3(config-if)#switchport trunk allowed
vlnswitchport trunk allowed vlan all
^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config-if)#switchport trunk allowed vlan
all
msk-ansusenko-sw-3(config-if)#no shutdown
```

Настройка транкового режима на порту Fa0/23



```
msk-ansusenko-sw-01
Physical  Config  CLI  Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to down

msk-ansusenko-sw-01(config-if)#no switchport mode trunk
msk-ansusenko-sw-01(config-if)#no switchport trunk allowed vlan
msk-ansusenko-sw-01(config-if)#switchport mode access
msk-ansusenko-sw-01(config-if)#no shutdown

msk-ansusenko-sw-01(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
GigabitEthernet0/2 VLAN1.

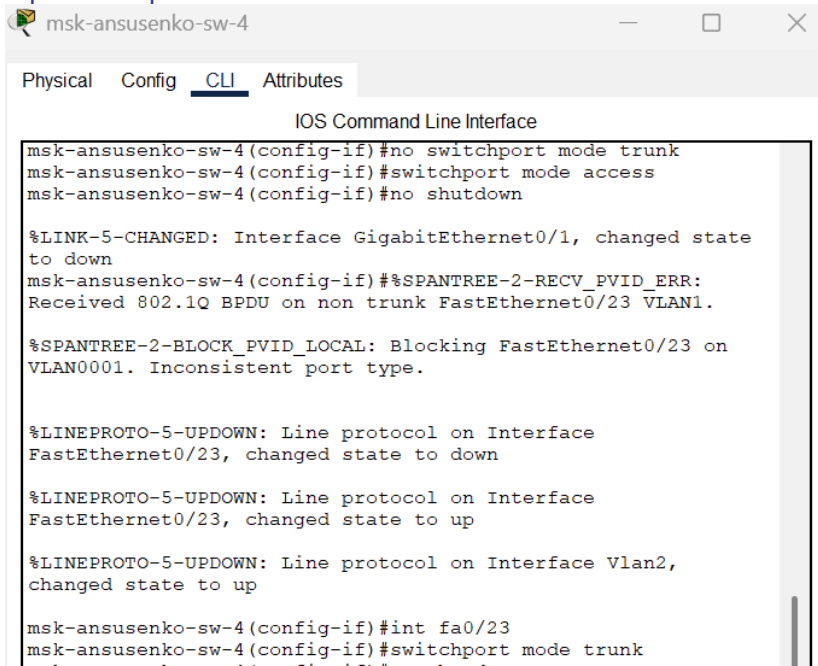
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/2 on VLAN0001.
Inconsistent port type.

msk-ansusenko-sw-01(config-if)#int g0/2
msk-ansusenko-sw-01(config-if)#int fa0/23
msk-ansusenko-sw-01(config-if)#switchport mode trunk

msk-ansusenko-sw-01(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23
```

Настройка транков на msk-ansusenko-sw-4



The screenshot shows a terminal window for a network switch named 'msk-ansusenko-sw-4'. The window has a title bar with standard OS controls (minimize, maximize, close) and a tabbed interface with 'Physical', 'Config', 'CLI', and 'Attributes' tabs. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal text shows a sequence of commands and system messages: first, 'no switchport mode trunk' is entered; then, 'switchport mode access' is entered, which triggers a series of status messages including '%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to down', '%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/23 VLAN1.', '%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/23 on VLAN0001. Inconsistent port type.', and two instances of '%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down'. Finally, 'int fa0/23' is entered, followed by 'switchport mode trunk'.

```
msk-ansusenko-sw-4
msk-ansusenko-sw-4(config-if)#no switchport mode trunk
msk-ansusenko-sw-4(config-if)#switchport mode access
msk-ansusenko-sw-4(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state
to down
msk-ansusenko-sw-4(config-if)#%SPANTREE-2-RECV_PVID_ERR:
Received 802.1Q BPDU on non trunk FastEthernet0/23 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/23 on
VLAN0001. Inconsistent port type.

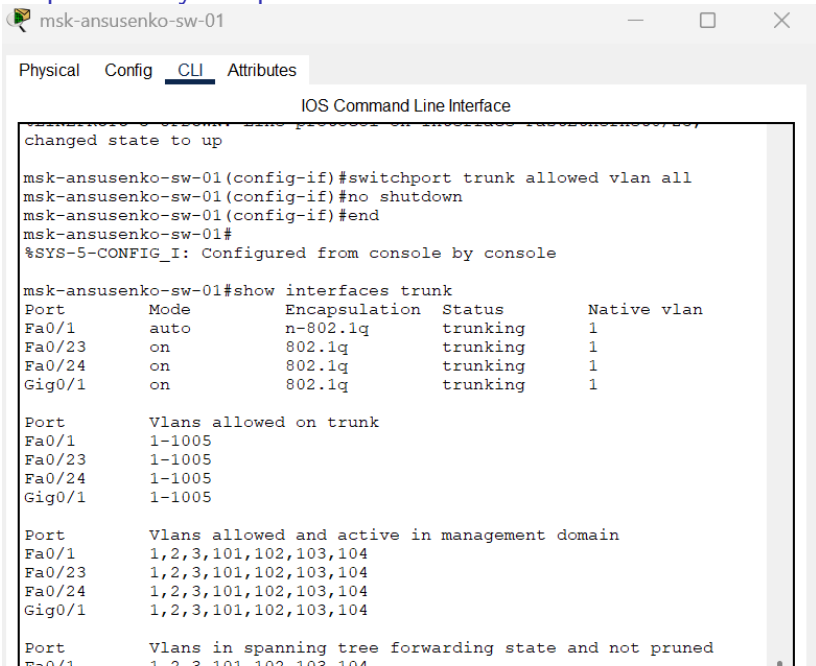
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2,
changed state to up

msk-ansusenko-sw-4(config-if)#int fa0/23
msk-ansusenko-sw-4(config-if)#switchport mode trunk
```


Проверка статуса транков на msk-ansusenko-sw-01



The screenshot shows a terminal window titled "msk-ansusenko-sw-01" with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the "IOS Command Line Interface".

The terminal output shows the following commands and results:

```
msk-ansusenko-sw-01(config-if)#switchport trunk allowed vlan all
msk-ansusenko-sw-01(config-if)#no shutdown
msk-ansusenko-sw-01(config-if)#end
msk-ansusenko-sw-01#
%SYS-5-CONFIG_I: Configured from console by console

msk-ansusenko-sw-01#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	auto	n-802.1q	trunking	1
Fa0/23	on	802.1q	trunking	1
Fa0/24	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1


```
Port          Vlans allowed on trunk
Fa0/1         1-1005
Fa0/23        1-1005
Fa0/24        1-1005
Gig0/1        1-1005
```



```
Port          Vlans allowed and active in management domain
Fa0/1         1,2,3,101,102,103,104
Fa0/23        1,2,3,101,102,103,104
Fa0/24        1,2,3,101,102,103,104
Gig0/1        1,2,3,101,102,103,104
```



```
Port          Vlans in spanning tree forwarding state and not pruned
Fa0/1         1,2,3,101,102,103,104
```

Проверка статуса транков на msk-ansusenko-sw-3

msk-ansusenko-sw-3

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-ansusenko-sw-3#show interfaces trunk
```

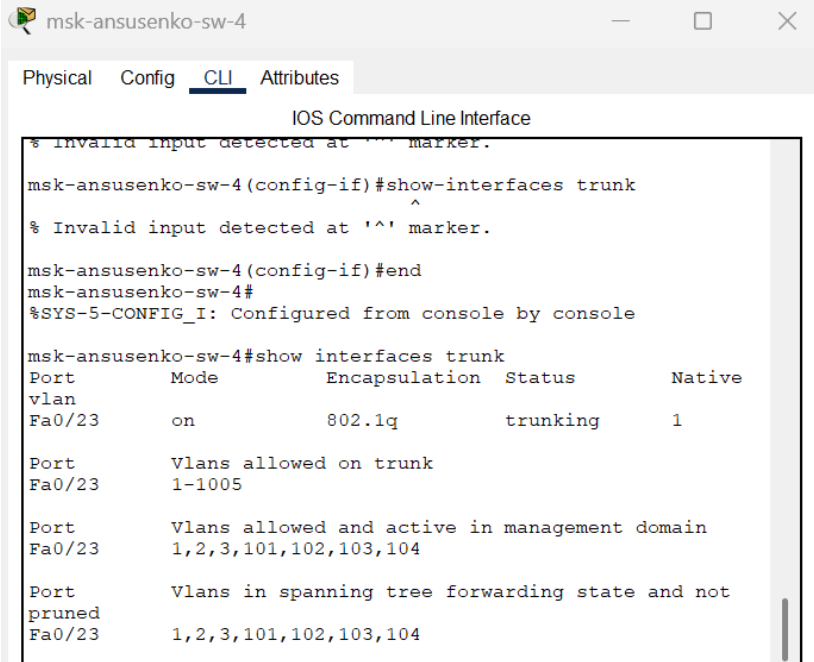
Port	Mode	Encapsulation	Status	Native
Gig0/1	on	802.1q	trunking	1
Gig0/2	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig0/1	1-1005
Gig0/2	1-1005

Port	Vlans allowed and active in management domain
Gig0/1	1,2,3,101,102,103,104
Gig0/2	1,2,3,101,102,103,104

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	1,2,3,101,102,103,104
Gig0/2	1,2,3,101,102,103,104

Проверка статуса транков на msk-ansusenko-sw-4



The screenshot shows a terminal window with the title bar 'msk-ansusenko-sw-4'. Inside the window, there are tabs for 'Physical', 'Config', 'CLI', and 'Attributes', with 'CLI' being the active tab. The main content area is titled 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
% Invalid input detected at '!' marker.

msk-ansusenko-sw-4(config-if)#show-interfaces trunk
      ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-4(config-if)#end
msk-ansusenko-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

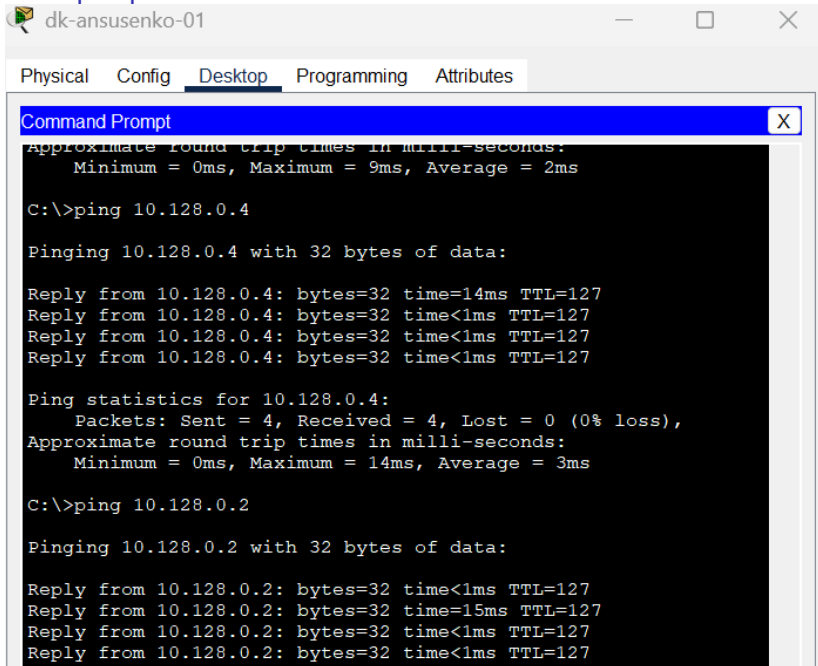
msk-ansusenko-sw-4#show interfaces trunk
Port      Mode           Encapsulation  Status        Native
vlan
Fa0/23    on              802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/23    1-1005

Port      Vlans allowed and active in management domain
Fa0/23    1,2,3,101,102,103,104

Port      Vlans in spanning tree forwarding state and not
pruned
Fa0/23    1,2,3,101,102,103,104
```

Пинг серверов с dk-ansusenko-01



```
dk-ansusenko-01
Physical  Config  Desktop  Programming  Attributes

Command Prompt [X]
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=14ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time=15ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
```

Пинг почтового сервера через коммутатор

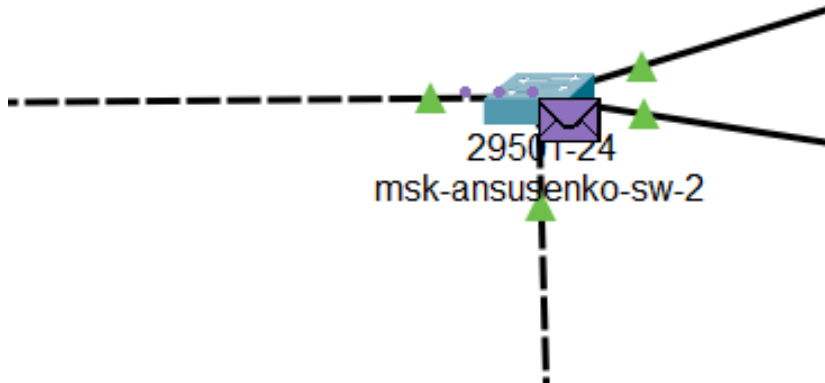


Рис.: Пинг почтового сервера через коммутатор msk-ansusenko-sw-2.

Состояние STP для VLAN 3 на msk-ansusenko-sw-2

msk-ansusenko-sw-2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!

User Access Verification

Password:

msk-ansusenko-sw-2>show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0001.4272.C298
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32771  (priority 32768 sys-id-ext 3)
             Address     0001.4272.C298
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface      Role  Sts  Cost           Prio.Nbr  Type
-----
Fa0/1          Desg  FWD  19             128.1     P2p
Gi0/2          Desg  FWD  4              128.26    P2p
Gi0/1          Desg  FWD  4              128.25    P2p
```

Настройка корневого коммутатора

msk-ansusenko-sw-01

Physical Config CLI Attributes

IOS Command Line Interface

```
Fa0/9, Fa0/10
Fa0/12, Fa0/13, Fa0/14
Fa0/16, Fa0/17, Fa0/18
Fa0/20, Fa0/21, Fa0/22
2    management    active
3    servers        active
101  dk             active
102  departaments   active
103  adm            active
104  other          active
1002 fddi-default    active
1003 token-ring-default
1004 fddinet-default active
1005 trnet-default   active
Fa0/11,
Fa0/15,
Fa0/19,
Gig0/2
Fa0/6
```

msk-ansusenko-sw-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

msk-ansusenko-sw-01(config)#spanning tree vlan 3 root primary

% Invalid input detected at '^' marker.

msk-ansusenko-sw-01(config)#spanning-tree vlan 3 root primary

Состояние для VLAN 3 на корневом коммутаторе

```
msh-ansusenko-sw-01#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID      Priority      24579
                Address      00D0.BCEE.A8B6
                This bridge is the root
                Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority      24579  (priority 24576 sys-id-ext 3)
                Address      00D0.BCEE.A8B6
                Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1    Shr
Gi0/1          Desg FWD 4         128.25   P2p
Fa0/24         Desg FWD 19        128.24   P2p
Fa0/23         Desg FWD 19        128.23   P2p

msh-ansusenko-sw-01#
```

Рис.: Состояние для VLAN 3 на корневом коммутаторе.

Движение пакетов до mail

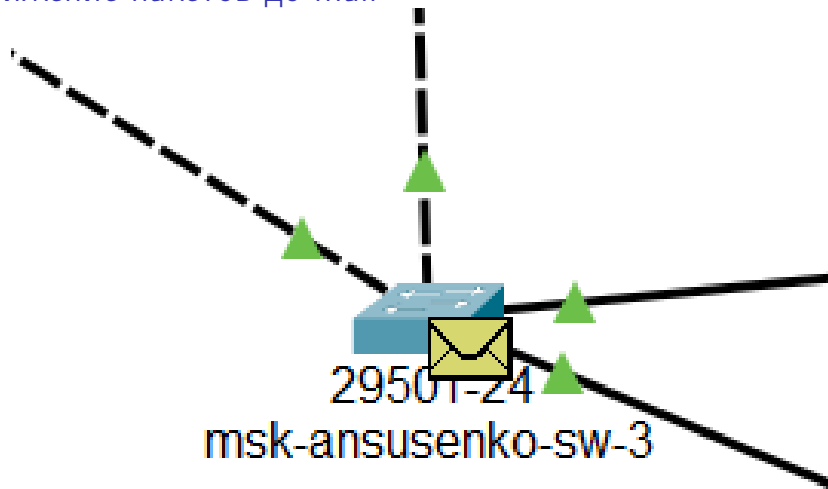


Рис.: Движение пакетов ICMP от dk-ansusenko-01 до mail через

Движение пакетов до web

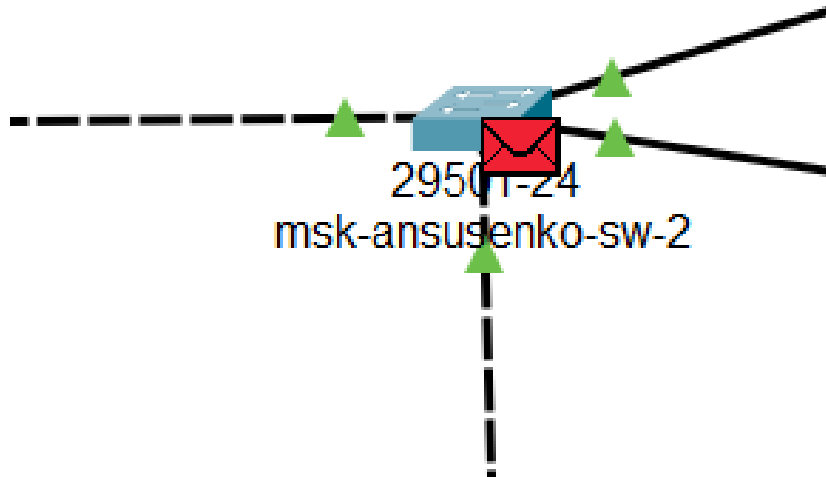
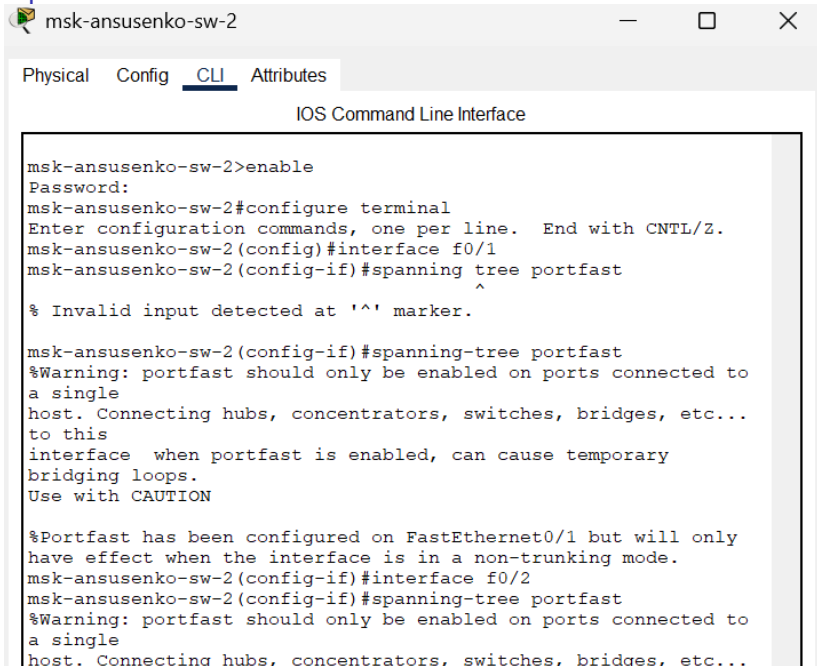


Рис.: Движение пакетов ICMP от dk-ansusenko-01 до web через msk-ansusenko-sw-01 и msk-ansusenko-sw-2.

Настройка Portfast на msk-ansusenko-sw-2

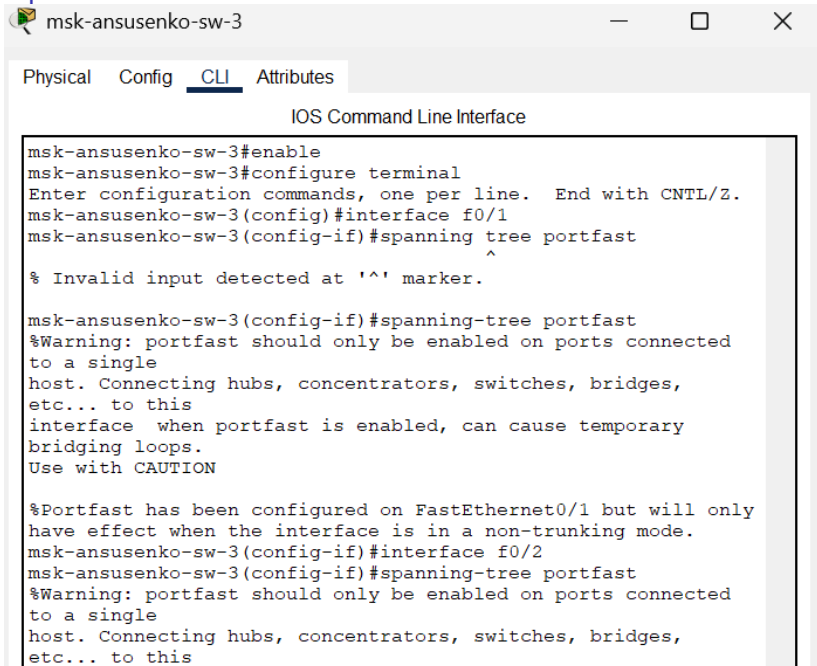


```
msk-ansusenko-sw-2>enable
Password:
msk-ansusenko-sw-2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-ansusenko-sw-2(config)#interface f0/1
msk-ansusenko-sw-2(config-if)#spanning tree portfast
                                     ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to
a single
host. Connecting hubs, concentrators, switches, bridges, etc...
to this
interface when portfast is enabled, can cause temporary
bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-ansusenko-sw-2(config-if)#interface f0/2
msk-ansusenko-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to
a single
host. Connecting hubs, concentrators, switches, bridges, etc...
```

Настройка Portfast на msk-ansusenko-sw-3

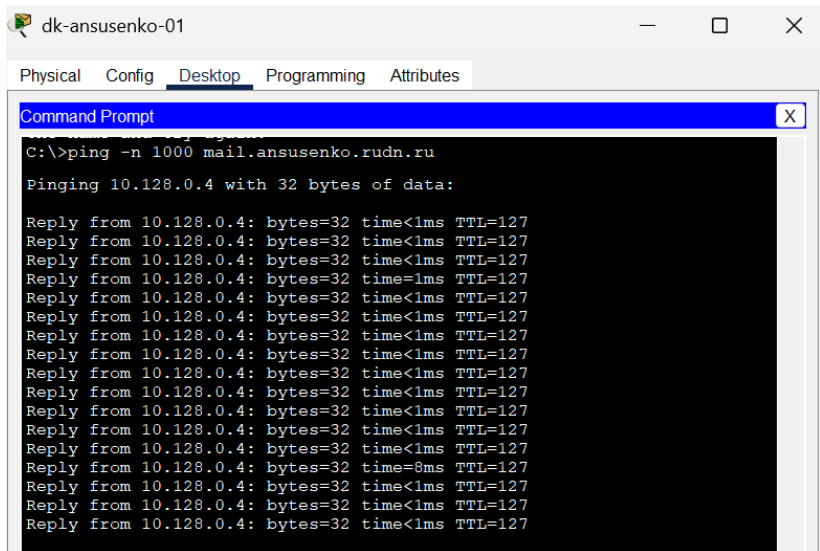


```
msk-ansusenko-sw-3#enable
msk-ansusenko-sw-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-ansusenko-sw-3(config)#interface f0/1
msk-ansusenko-sw-3(config-if)#spanning tree portfast
^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected
to a single
host. Connecting hubs, concentrators, switches, bridges,
etc... to this
interface when portfast is enabled, can cause temporary
bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-ansusenko-sw-3(config-if)#interface f0/2
msk-ansusenko-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected
to a single
host. Connecting hubs, concentrators, switches, bridges,
etc... to this
```

Длительный пинг до почтового сервера (с разрывом)



The screenshot shows a Windows desktop environment. At the top, there is a taskbar with a single icon labeled 'dk-ansusenko-01'. Below the taskbar, there are several tabs: 'Physical', 'Config', 'Desktop' (which is selected), 'Programming', and 'Attributes'. The 'Desktop' tab contains a 'Command Prompt' window. The Command Prompt has a title bar with 'X' and a blue background. The text inside the Command Prompt is as follows:

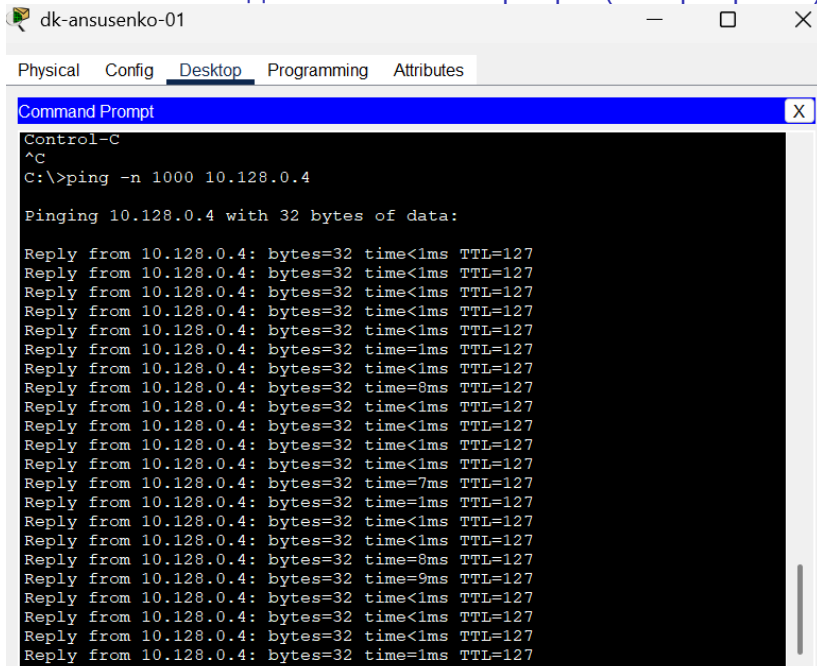
```
C:\>ping -n 1000 mail.ansusenko.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=8ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
```

Рис.: Длительный пинг до почтового сервера с разрывом соединения.

Длительный пинг до почтового сервера (без разрыва)



```
Control-C  
^C  
C:\>ping -n 1000 10.128.0.4  
  
Pinging 10.128.0.4 with 32 bytes of data:  
  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=8ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=7ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=8ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=9ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127  
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
```

Логическая схема с агрегированным соединением

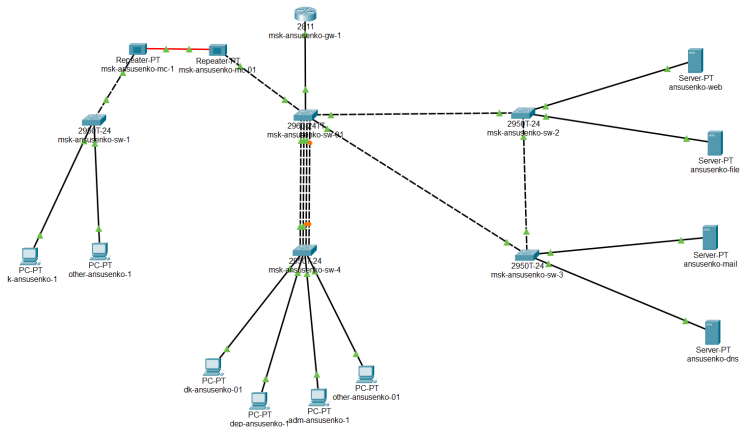
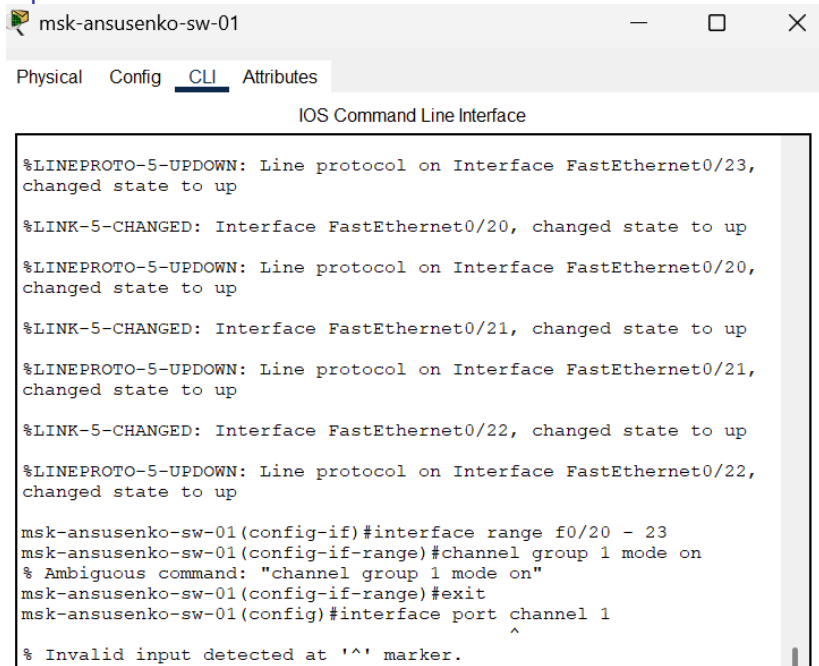


Рис.: Логическая схема локальной сети с агрегированным соединением.

Настройка EtherChannel на msk-ansusenko-sw-01



```
msk-ansusenko-sw-01
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/20,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to up

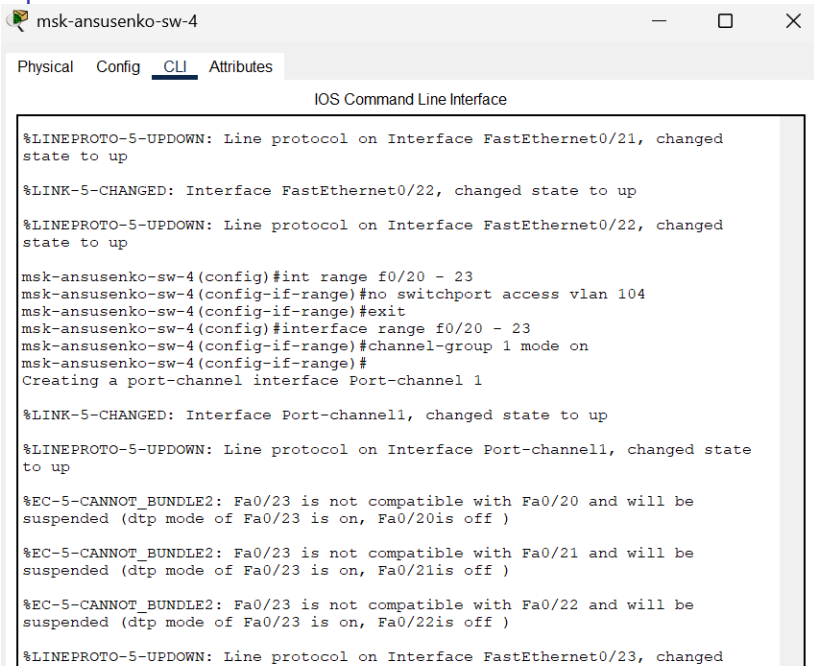
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22,
changed state to up

msk-ansusenko-sw-01(config-if)#interface range f0/20 - 23
msk-ansusenko-sw-01(config-if-range)#channel group 1 mode on
% Ambiguous command: "channel group 1 mode on"
msk-ansusenko-sw-01(config-if-range)#exit
msk-ansusenko-sw-01(config)#interface port channel 1
^
% Invalid input detected at '^' marker.
```


Настройка EtherChannel на msk-ansusenko-sw-4



The screenshot shows a network device's Command Line Interface (CLI) window. The window title is "msk-ansusenko-sw-4". At the top, there are tabs for "Physical", "Config", "CLI" (which is selected), and "Attributes". Below the tabs, the text "IOS Command Line Interface" is displayed. The main area of the window contains a series of status messages and configuration commands. The status messages indicate that line protocols and interfaces (FastEthernet0/21, FastEthernet0/22, and Port-channel1) have changed state to up. The configuration commands show the creation of a port-channel interface named "Port-channel 1" by aggregating interfaces "f0/20" and "f0/23". The commands also show the configuration of the port-channel interface, including setting the mode to "channel-group 1" and the access VLAN to "104". Finally, the status messages indicate that the line protocol on the port-channel interface has changed state to up.

```
msk-ansusenko-sw-4
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed
state to up

msk-ansusenko-sw-4(config)#int range f0/20 - 23
msk-ansusenko-sw-4(config-if-range)#no switchport access vlan 104
msk-ansusenko-sw-4(config-if-range)#exit
msk-ansusenko-sw-4(config)#interface range f0/20 - 23
msk-ansusenko-sw-4(config-if-range)#channel-group 1 mode on
msk-ansusenko-sw-4(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state
to up

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be
suspended (dtp mode of Fa0/23 is on, Fa0/20is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be
suspended (dtp mode of Fa0/23 is on, Fa0/21is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be
suspended (dtp mode of Fa0/23 is on, Fa0/22is off )

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed
```

Контрольный вопрос №1

Какую информацию можно получить, воспользовавшись командой определения состояния протокола STP для VLAN (на корневом и не на корневом устройстве)?

Приведите примеры.

Команда `show spanning-tree vlan <id>` позволяет получить:

- ▶ Приоритет и MAC-адрес корневого устройства (Root ID)
- ▶ Приоритет и MAC-адрес текущего моста (Bridge ID)
- ▶ Роли портов (Root, Designated, Alternate)
- ▶ Состояние портов (FWD, BLK, LSN, LRN)
- ▶ Стоимость портов (Cost) и приоритеты портов (Prio.Nbr)

Контрольный вопрос №1 (пример вывода)

На корневом устройстве (msk-ansusenko-sw-01):

```
Root ID      Priority    24579
              Address     00D0.BCEE.A8B6
              This bridge is the root
Bridge ID    Priority    24579
              Address     00D0.BCEE.A8B6

Interface      Role Sts Cost
Fa0/1          Desg FWD 19
Gi0/1          Desg FWD 4
Gi0/2          Desg FWD 1
```

Контрольный вопрос №2

При помощи какой команды можно узнать, в каком режиме (STP или Rapid PVST+) работает устройство?

Команда `show spanning-tree vlan <id>`. В первой строке вывода указывается используемый протокол:

- ▶ `Spanning tree enabled protocol ieee` - классический STP (802.1d)
- ▶ `Spanning tree enabled protocol rstp` - Rapid STP (802.1w)
- ▶ `Spanning tree enabled protocol mstp` - Multiple STP (802.1s)

Контрольный вопрос №2 (пример вывода)

Пример вывода для STP:

VLAN0003

Spanning tree enabled protocol ieee

Root ID	Priority	32771
	Address	0001.4272.C298
This bridge is the root		

Пример вывода для Rapid PVST+:

VLAN0003

Spanning tree enabled protocol rstp

Root ID	Priority	24579
	Address	00D0.BCEE.A8B6

Контрольный вопрос №3

Для чего и в каких случаях нужно настраивать режим Portfast?

Portfast настраивается на access-портах коммутатора, к которым подключены конечные устройства (компьютеры, серверы, принтеры).

позволяет порту сразу переходить в режим Forwarding, минуя стадии Listening и Learning ($15 + 15 = 30$ секунд).

Когда использовать:

- ▶ На портах, подключенных к одному хосту
- ▶ На портах, где не может образоваться петля
- ▶ Для ускорения загрузки ОС по сети (PXE)

Контрольный вопрос №4

В чем состоит принцип работы агрегированного интерфейса? Для чего он используется?

EtherChannel объединяет несколько физических портов в один логический канал (Port-channel).

Особенности:

- ▶ Трафик распределяется между физическими портами по алгоритму хэширования (MAC-адреса, IP-адреса, порты TCP/UDP)
- ▶ Все порты в группе должны иметь одинаковые настройки (скорость, дуплекс, VLAN, trunk)

Для чего используется:

- ▶ Увеличение пропускной способности (агрегирование полосы)
- ▶ Отказоустойчивость (при выходе порта из строя трафик идёт по остальным)
- ▶ Балансировка нагрузки между портами

Контрольный вопрос №5

В чём принципиальные отличия при использовании протоколов LACP, PAgP и статического агрегирования?

Характеристика	LACP (802.3ad)	PAgP (Cisco)	Static (on)
Тип	Открытый стандарт	Проприетарный Cisco	Без про
Совместимость	Разные вендоры	Только Cisco	Любое дование
Обнаружение ошибок	Да	Да	Нет
Режимы	active/passive	desirable/auto	on

Вывод: LACP рекомендуется для мультивендорных сетей, PAgP - для сетей только Cisco, Static - для простых конфигураций без необходимости протокольного согласования.

Контрольный вопрос №6

При помощи каких команд можно узнать состояние агрегированного канала EtherChannel?

Основные команды для проверки EtherChannel:

Краткая информация о всех EtherChannel
`show etherchannel summary`

Детальная информация о Port-channel
`show etherchannel port-channel`

Статус логического интерфейса
`show interfaces port-channel <номер>`

Статус физических портов в группе
`show interfaces status`

Информация о балансировке нагрузки
`show etherchannel load-balance`

Контрольный вопрос №6 (пример вывода)

Пример вывода `show etherchannel summary`:

Flags: D - down P - in port-channel
I - stand-alone s - suspended

Group	Port-channel	Protocol	Ports
-------	--------------	----------	-------

1	Po1(SU)	-	Fa0/20(P) Fa0/21(P) Fa0/22(P) Fa0/23(P)
---	---------	---	--

Расшифровка: SU - Port-channel работает (S - switched, U - up), все порты в группе активны (P).

Выводы

- ▶ В ходе выполнения лабораторной работы были изучены протоколы STP и Rapid PVST+, а также технология агрегирования каналов EtherChannel. Настроено резервное соединение между коммутаторами, обеспечена отказоустойчивость сети, активирован режим Portfast для серверных портов. Проверена работоспособность ICMP-запросов в режиме симуляции. Все поставленные задачи выполнены.